

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
print("Libraries imported successfully")
```

Libraries imported successfully

```
In [4]: df=pd.read_csv(r"C:\Users\ELCOT\Downloads\titanic\train.csv")
df.head()
```

```
Out[4]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450

Titanic Dataset - Exploratory Data Analysis

Objective: Extract insights using visual and statistical exploration

Dataset: Titanic Training Data (891 passengers, 12 columns)

```
In [6]: print("Data shape:", df.shape)
print("\n")
```

```
print("Dataset info")
print(df.info())
```

Data shape: (891, 12)

```
Dataset info
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass          891 non-null    int64
3   Name             891 non-null    str
4   Sex             891 non-null    str
5   Age             714 non-null    float64
6   SibSp           891 non-null    int64
7   Parch           891 non-null    int64
8   Ticket          891 non-null    str
9   Fare           891 non-null    float64
10  Cabin           204 non-null    str
11  Embarked        889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
```

```
In [7]: print("Statistical Summary")
df.describe()
```

Statistical Summary

```
Out[7]:
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000

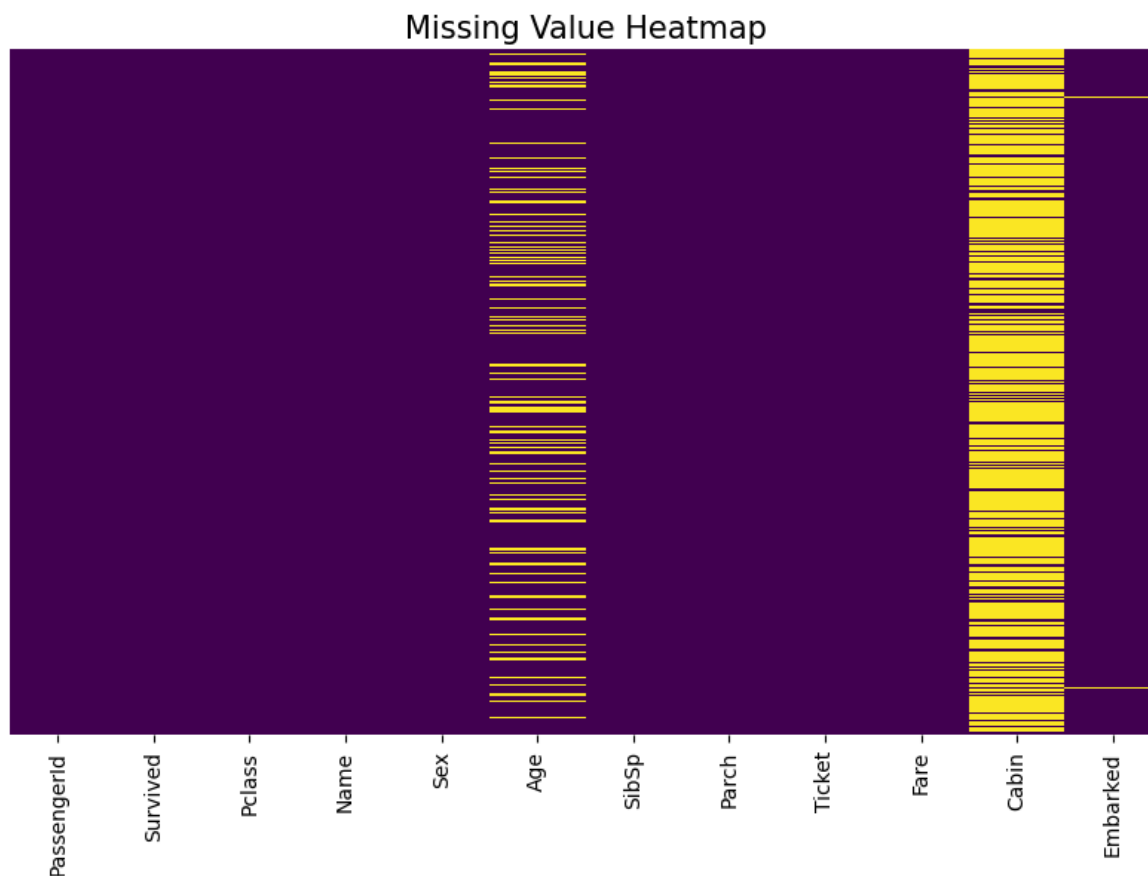
```
In [8]: print("Missing values in each column")
print(df.isnull().sum())
print("\n")
print("\nMissing Value Percentage")
print(round(df.isnull().sum()/len(df)* 100,2))
```

```
Missing values in each column
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

```
Missing Value Percentage
PassengerId      0.00
Survived          0.00
Pclass           0.00
Name             0.00
Sex              0.00
Age             19.87
SibSp            0.00
Parch            0.00
Ticket           0.00
Fare             0.00
Cabin           77.10
Embarked         0.22
dtype: float64
```

Missing Value Heatmap

```
In [9]: plt.figure(figsize=(10,6))
sns.heatmap(df.isnull(),yticklabels=False,cbar=False,cmap='viridis')
plt.title('Missing Value Heatmap',fontsize=15)
plt.show()
```



```
In [13]: df['Age']=pd.to_numeric(df['Age'],errors='coerce')
df['Age']=df['Age'].fillna(df['Age'].median())
df['Embarked']=df['Embarked'].fillna(df['Embarked'].mode()[0])
df=df.drop(columns=['Cabin'])
print("Missing values after cleaning:")
print(df.isnull().sum())
```

Missing values after cleaning:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

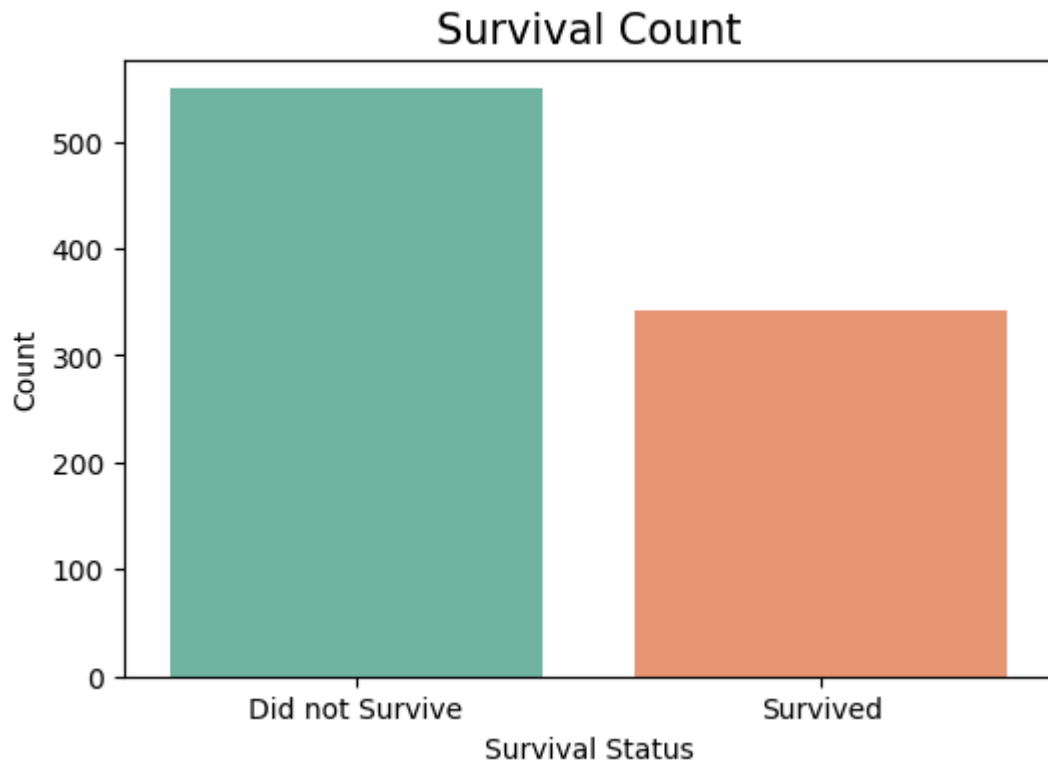
dtype: int64

Univariate Analysis

Survival Count

```
In [32]: plt.figure(figsize=(6,4))
sns.countplot(x='Survived',data=df,palette='Set2')
plt.title('Survival Count',fontsize=15)
plt.xticks([0,1],['Did not Survive','Survived'])
plt.xlabel('Survival Status')
```

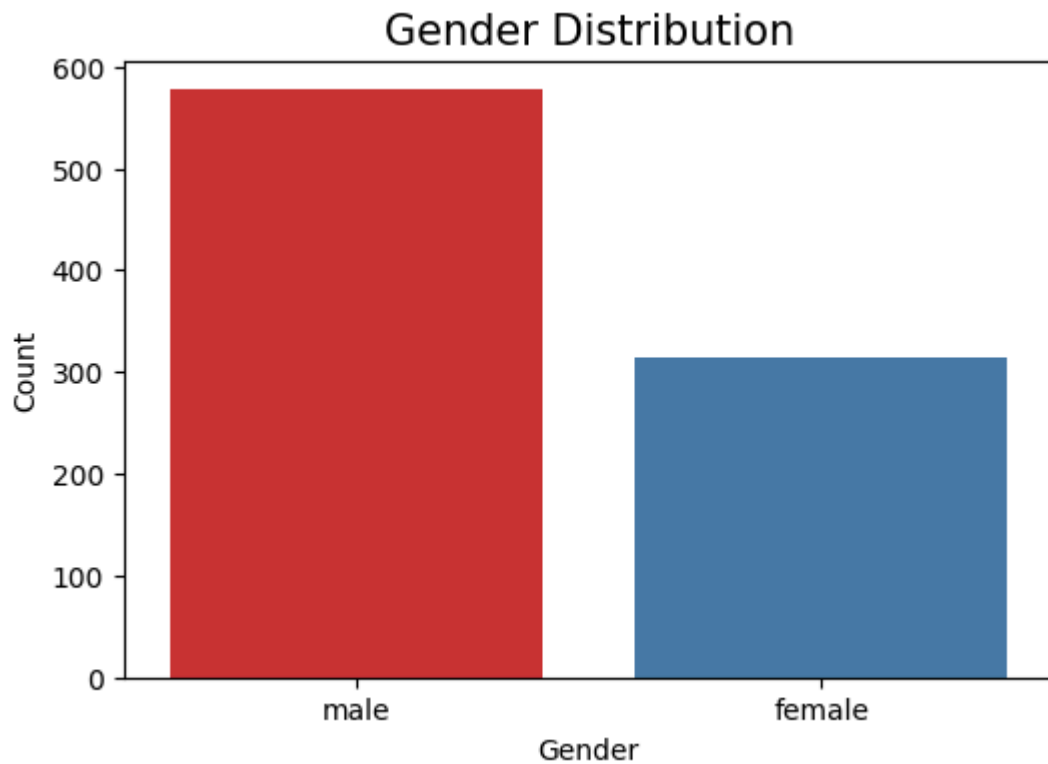
```
plt.ylabel('Count')
plt.show()
print("Survival Value Counts:")
print(df['Survived'].value_counts())
```



Survival Value Counts:
Survived
0 549
1 342
Name: count, dtype: int64

Gender Distribution

```
In [31]: plt.figure(figsize=(6,4))
sns.countplot(x='Sex',data=df,palette='Set1')
plt.title('Gender Distribution',fontsize=15)
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
print("Gender Value Counts:")
print(df['Sex'].value_counts())
```



Gender Value Counts:

Sex

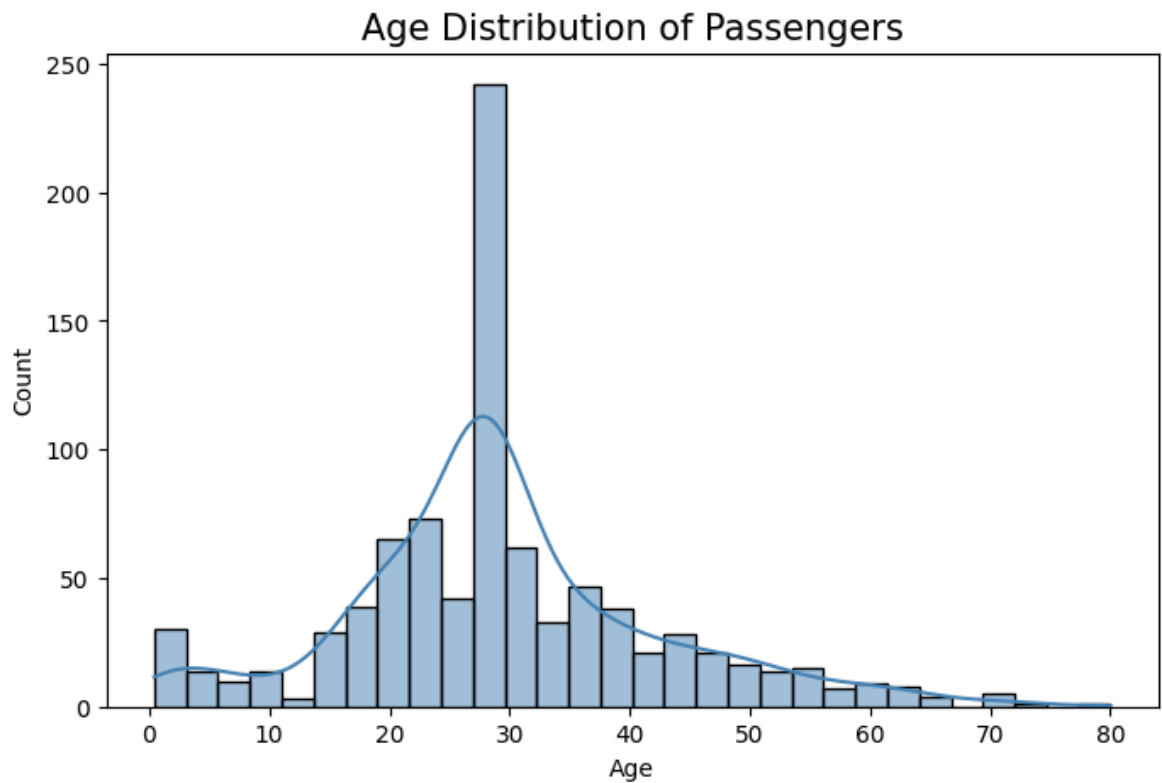
male 577

female 314

Name: count, dtype: int64

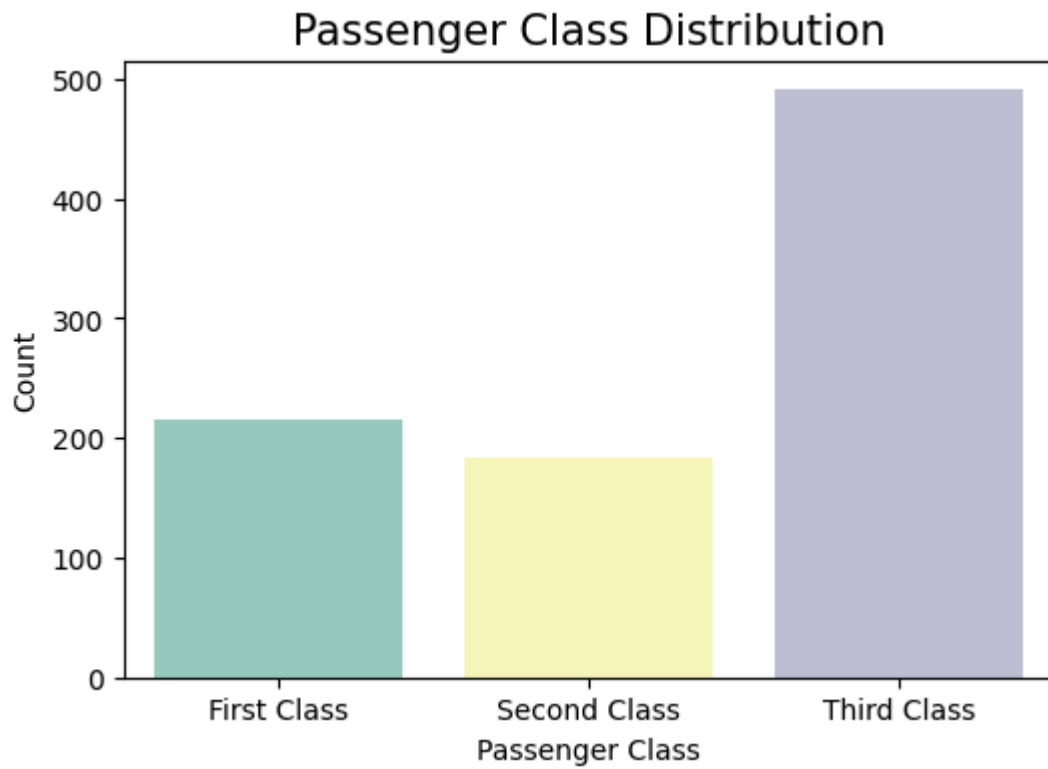
Age Distribution — Histogram

```
In [19]: plt.figure(figsize=(8,5))
sns.histplot(df['Age'],bins=30,kde=True,color='steelblue')
plt.title('Age Distribution of Passengers',fontsize=15)
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()
```



Passenger Class Distribution

```
In [21]: plt.figure(figsize=(6,4))
sns.countplot(x='Pclass',data=df,palette='Set3')
plt.title('Passenger Class Distribution',fontsize=15)
plt.xlabel('Passenger Class')
plt.ylabel('Count')
plt.xticks([0,1,2],['First Class','Second Class','Third Class'])
plt.show()
print("Passenger Class Value Counts:")
print(df['Pclass'].value_counts())
```



Passenger Class Value Counts:

Pclass

3 491

1 216

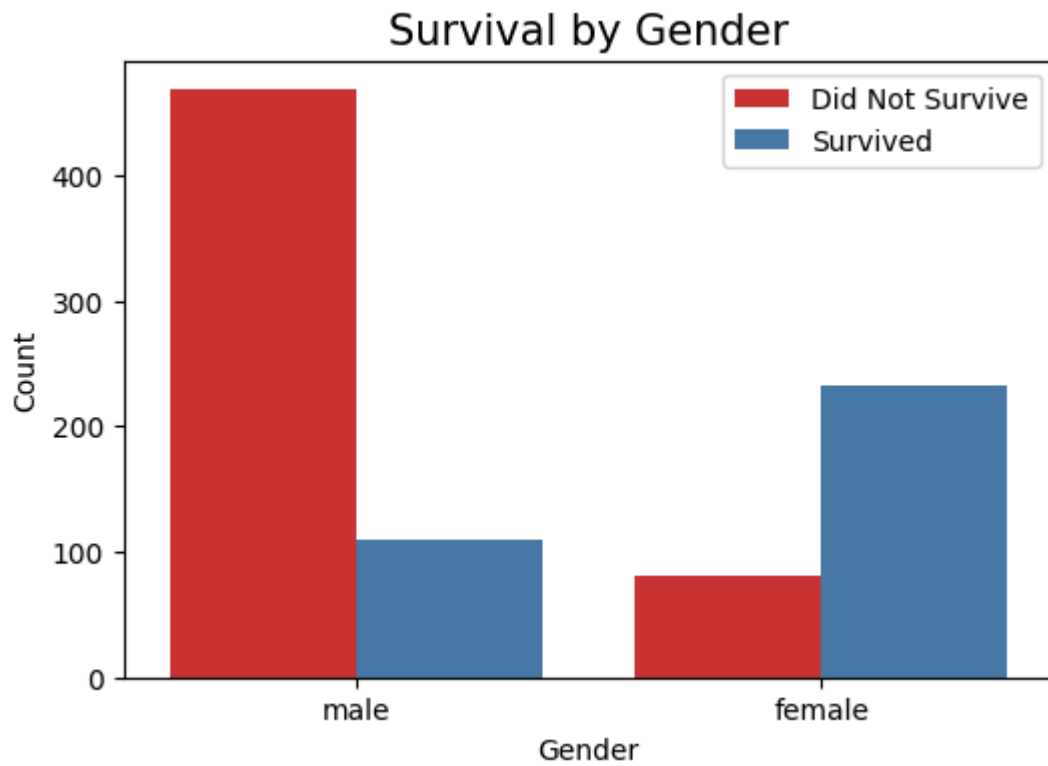
2 184

Name: count, dtype: int64

Bivariate Analysis

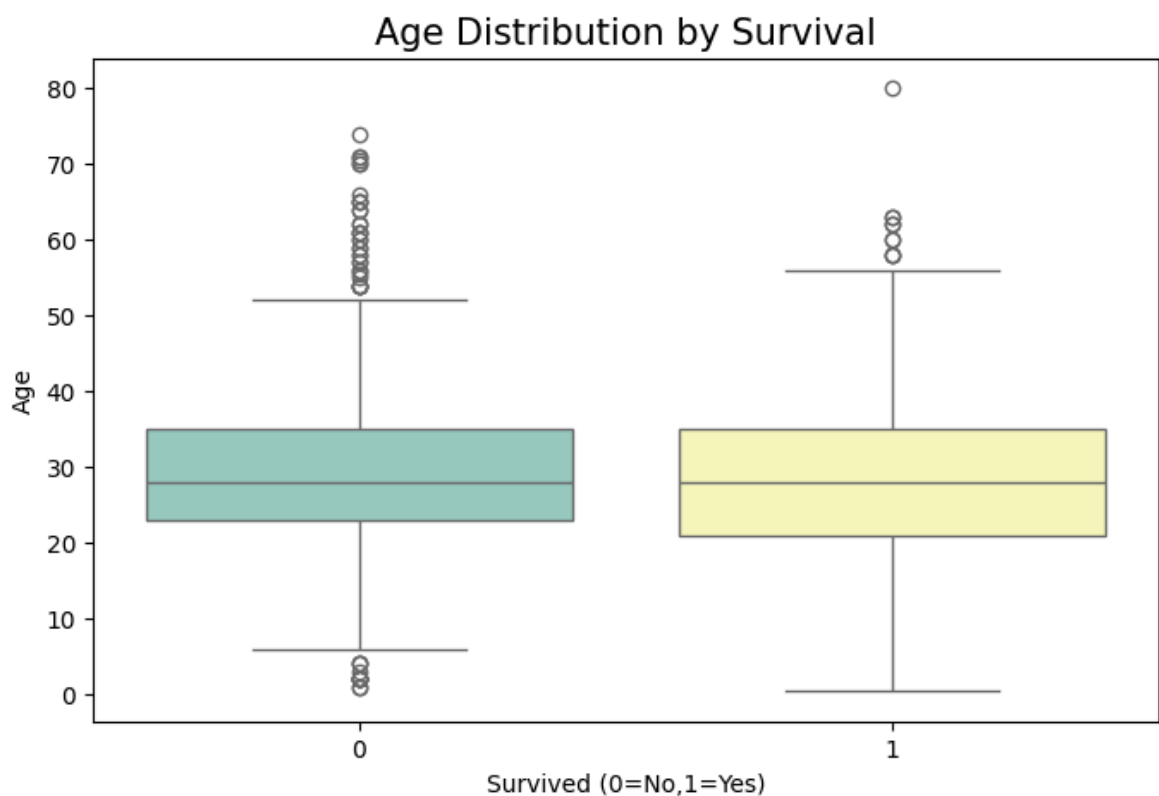
Survival by Gender

```
In [22]: plt.figure(figsize=(6,4))
sns.countplot(x='Sex',hue='Survived',data=df,palette='Set1')
plt.title('Survival by Gender',fontsize=15)
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['Did Not Survive','Survived'])
plt.show()
```



Age vs Survival Boxplot

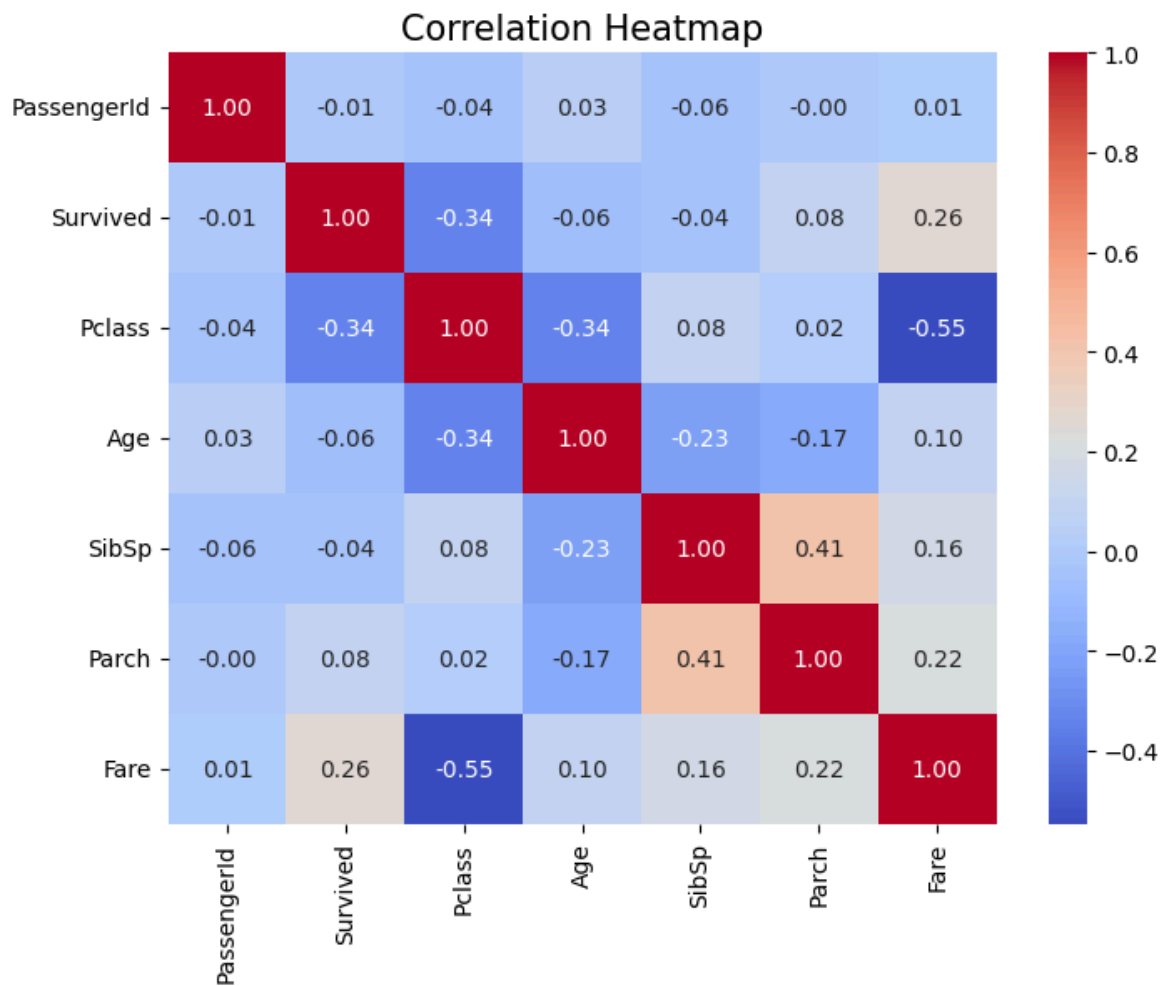
```
In [27]: plt.figure(figsize=(8,5))
sns.boxplot(x='Survived',y='Age',data=df,palette='Set3')
plt.title('Age Distribution by Survival', fontsize=15)
plt.xlabel('Survived (0=No,1=Yes)')
plt.ylabel('Age')
plt.show()
```



Multivariate Analysis

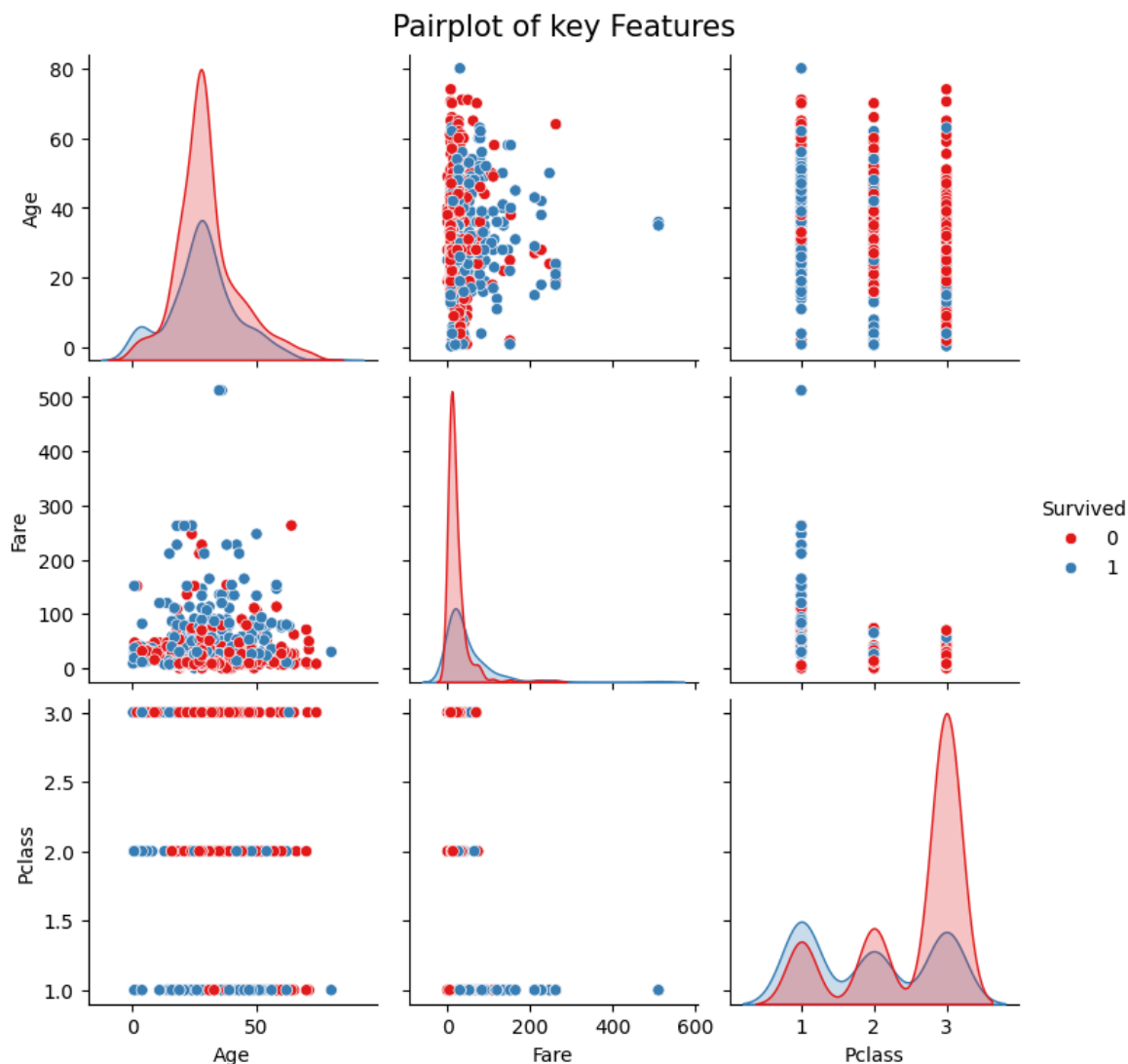
Correlation Heatmap

```
In [28]: plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True),annot=True,cmap='coolwarm',fmt='.2')
plt.title('Correlation Heatmap',fontsize=15)
plt.show()
```



Pairplot

```
In [29]: sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']],hue='Survived',palett
plt.suptitle('Pairplot of key Features',y=1.02,fontsize=15)
plt.show()
```



Key Findings and Summary

Key Insights from Titanic EDA

1. Survival Rate

Only 38% of passengers survived (342 out of 891 passengers). Majority of passengers did not survive the disaster.

2. Gender Impact

Women had a much higher survival rate than men. The Women and Children First policy is clearly visible in the data. Most male passengers did not survive while most female passengers survived.

3. Passenger Class Impact

First class passengers survived more than second and third class. Third class had the most passengers but the lowest survival rate. Wealthier passengers had better access to lifeboats.

4. Age Distribution

Most passengers were between 20 and 35 years old. Age alone was not a strong predictor of survival. Both survivors and non survivors had similar median age of around 28 to 30.

5. Fare and Class Correlation

Higher fare is strongly linked to higher passenger class with a correlation of -0.55. Passengers who paid higher fares had better survival chances. Fare and Survival show a positive correlation of 0.26.

6. Missing Data Handling

Age column had 19.87 percent missing values filled with median age. Cabin column had 77.10 percent missing values and was dropped entirely. Embarked column had 2 missing values filled with mode value.

Conclusion

The most important factors affecting survival on the Titanic were Gender, Passenger Class, and Fare paid. Social status and gender played a significant role in determining who survived the disaster. This EDA successfully identified key patterns, trends, and relationships within the Titanic dataset.

In []: